

CHAROTARUNIVERSITY OF SCIENCE & TECHNOLOGY
5th Semester of B. Pharm. Examination
University Backlog Theory Examination March 2018
PH303 Medicinal Chemistry-I [Old Syllabus]

Date: 29/03/2018, Thursday

Time: 10:00 a.m to 01:00 p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

- Q. 1** Attempt any **TWO** of the following:
- A** 1. Enumerate various factors affecting drug metabolism. Explain Phase-I metabolic reactions in detail. **10**
- B** 1. Define and classify prodrugs with suitable examples. Enumerate various applications of prodrug. **05**
2. What is CYP450? Give its function and importance. **05**
- C** 1. Write reaction and mechanism of the followings.(Any TWO) **10**
- a) Skraup synthesis
- b) Hantzsch synthesis
- c) Fischer indole synthesis
- Q. 2** Attempt any **TWO** of the following:
- A** **Write a Comments on: (ANY FOUR)** **10**
1. Pyridine is more basic than Pyrrole.
2. Impurities of Thiophene in benzene can be removed by cold sulphuric acid.
3. Electrophilic substitution takes place at 2nd and 5th position in Pyrrole.
4. Proton pump inhibitors act as pro-drug.
5. Furan is less aromatic than Thiophene.
- B** 1. Explain the chemistry, preparation and reactions of Pyrrole. **05**
2. Write a note on drug-receptor interaction. **05**
- C** 1. Write Paal-Knorr synthesis for Pyrrole with reaction mechanism. **05**
2. Give the structure of: **(ANY FIVE)** **05**
- (1) Quinoline (2) Triazole (3) Furan (4) Imidazole (5) Oxazole (6) Pepiridine

SECTION- II

Q. 3 Attempt any **TWO** of the following:

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| A | 1. Classify H ₁ receptor antagonist with suitable examples. | 05 |
| | 2. Antispasmodic agents | 05 |
| B | 1. Classify Reversible and Irreversible anticholinesterase. With the help of a diagram explain the mechanism of action of anticholinesterase drugs. | 10 |
| C | Give SAR and chemical mechanism of action of Antiasthmatic agents. Give scheme of synthesis of Salbutamol.. | 10 |

Q. 4 Attempt any **TWO** of the following:

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| A | Classify Antitussive and Expectorants. Give synthesis of Epinephrine. | 10 |
| B | Write short note (Classification, Structure, Uses) on : | 10 |
| | 1. Antacids | |
| | 2. Purgatives | |
| C | 1. Classify Adrenergic drugs and give synthesis of Norepinephrine. | 05 |
| | 2. Write a short note on Respiratory Stimulants. | 05 |
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